

LLM FINE-TUNING USING PERFORMANCE EFFICIENT FINE-TUNING (PEFT) METHODS

Thesis Presentation for Master of Science in Machine Learning & Artificial Intelligence

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Introduction

- Transformer Architecture
- Traditional Machine Learning Models
- Pre-Trained Language Models

Introduction cont...

- LLM Fine-Tuning
- Full Parameter Fine-Tuning and Challenges
- Parameter Efficient Fine-Tuning

Aims & Objectives

- To explore and understand language models
- To understand model Fine-Tuning using PEFT methods
 - *LoRA*
 - *QLoRA*
- To explore HuggingFace platform & open-source libraries
- To understand & use AWS SageMaker Notebook

Literature Review

■ Transformer Architecture

- *Attention Mechanism*
- *Transformer Model*

■ Pre-Trained Language Model & Parameter-Efficient Fine-Tuning

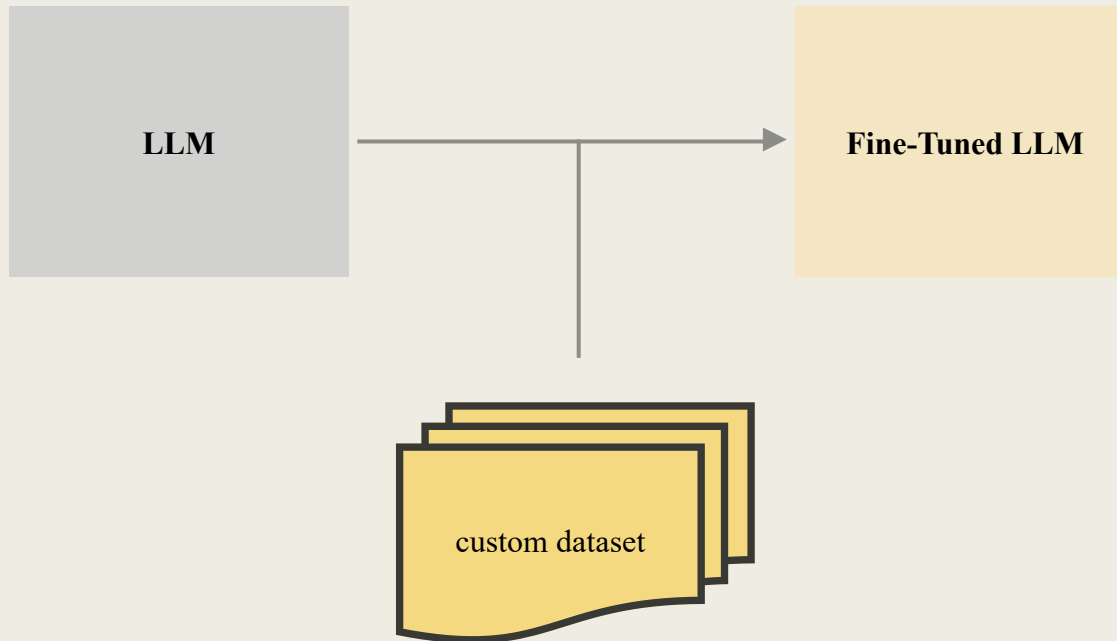
- *LoRA*
- *QLoRA*
- *Prefix Tuning*
- *Prompt Tuning*

Literature Review cont...

- Distributed PEFT
- Multiple Trained LoRA Composition
- LLM Hallucination
 - *Factuality*
 - *Faithfulness*

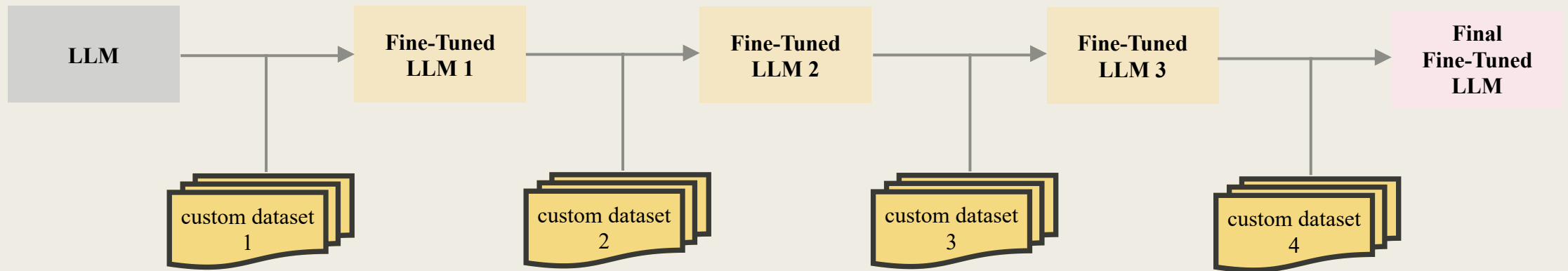
Methodology

Full Dataset Fine-Tuning (FDFT)



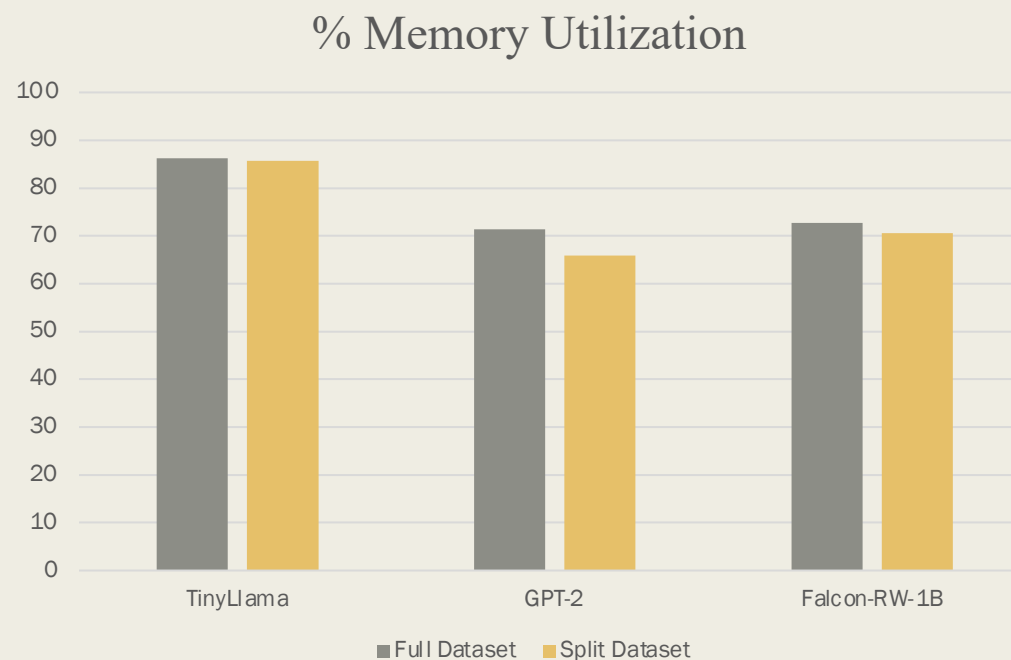
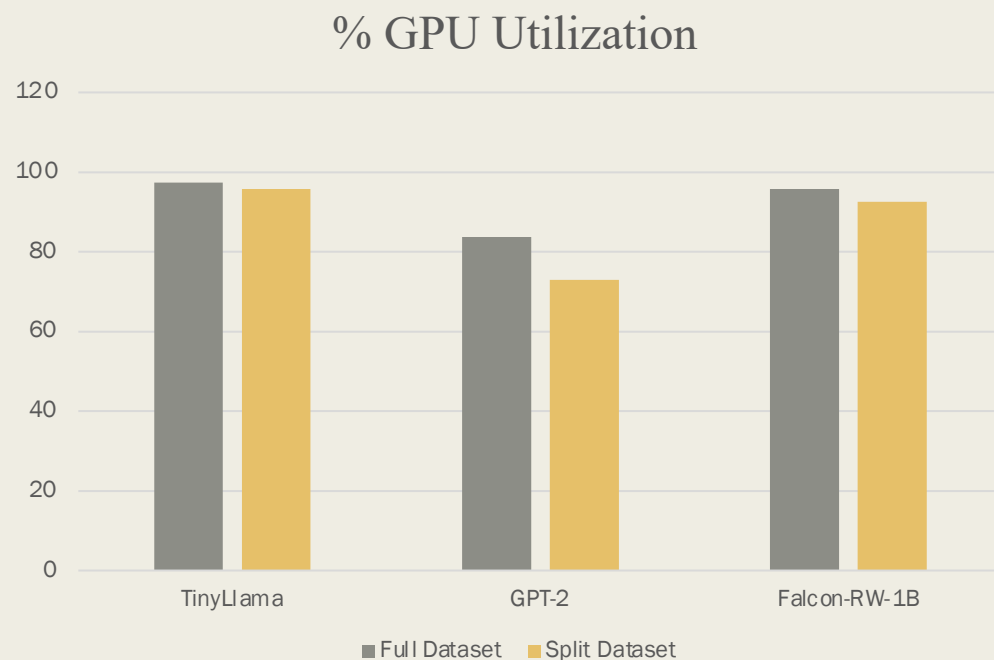
Methodology cont...

Split Dataset Fine-Tuning

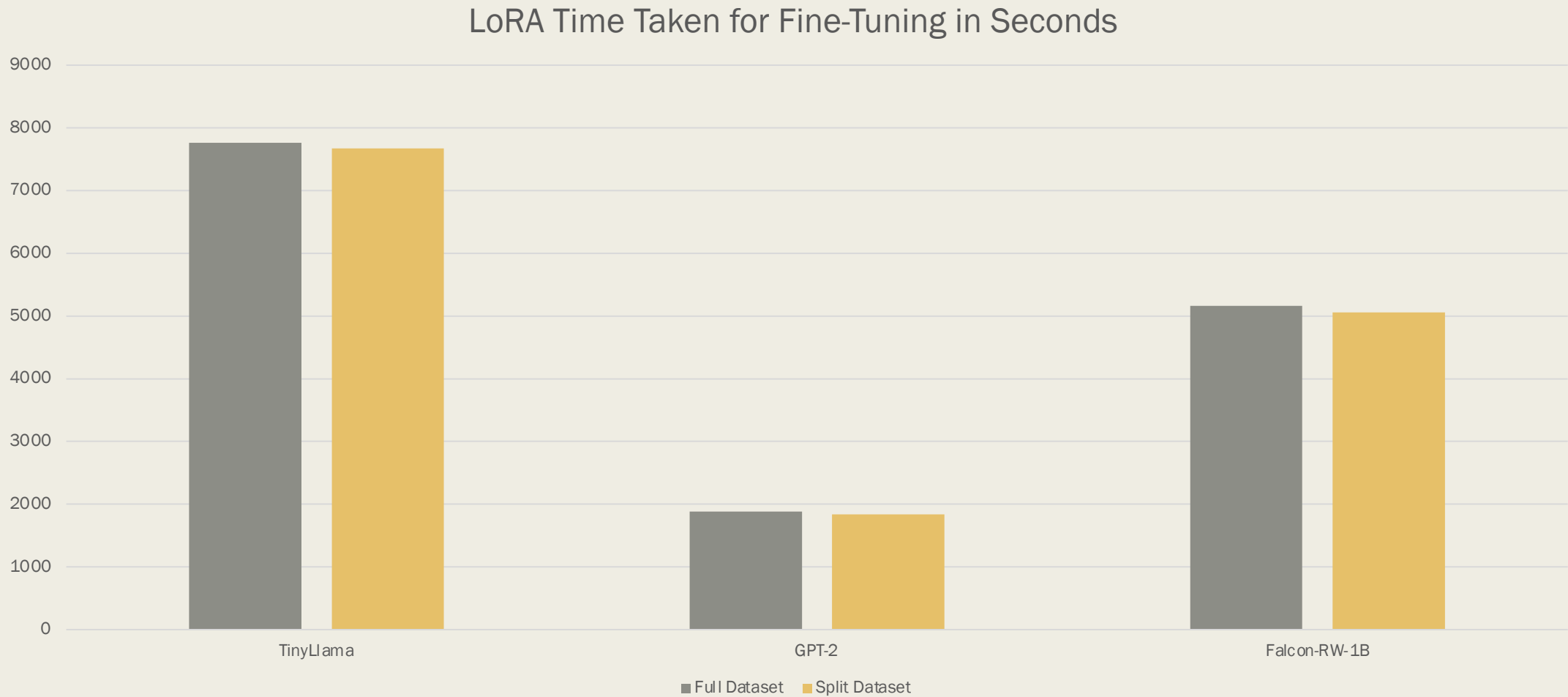


Results & Discussions

LORA Fine-Tuning Resource Utilization

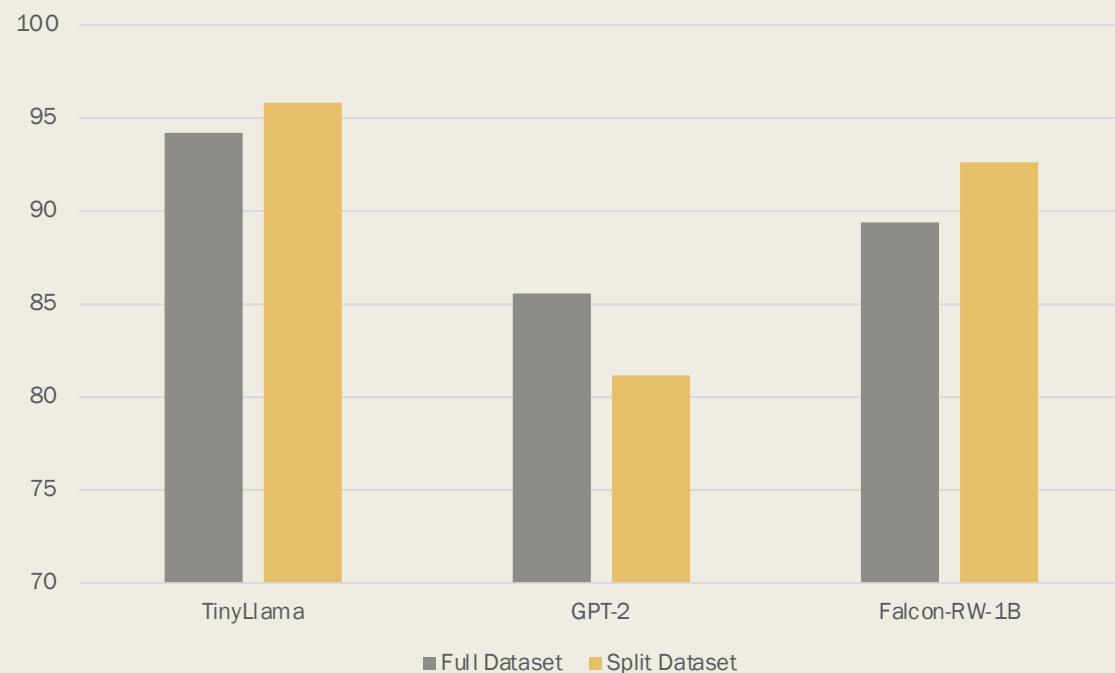


LoRA Fine-Tuning Time Taken

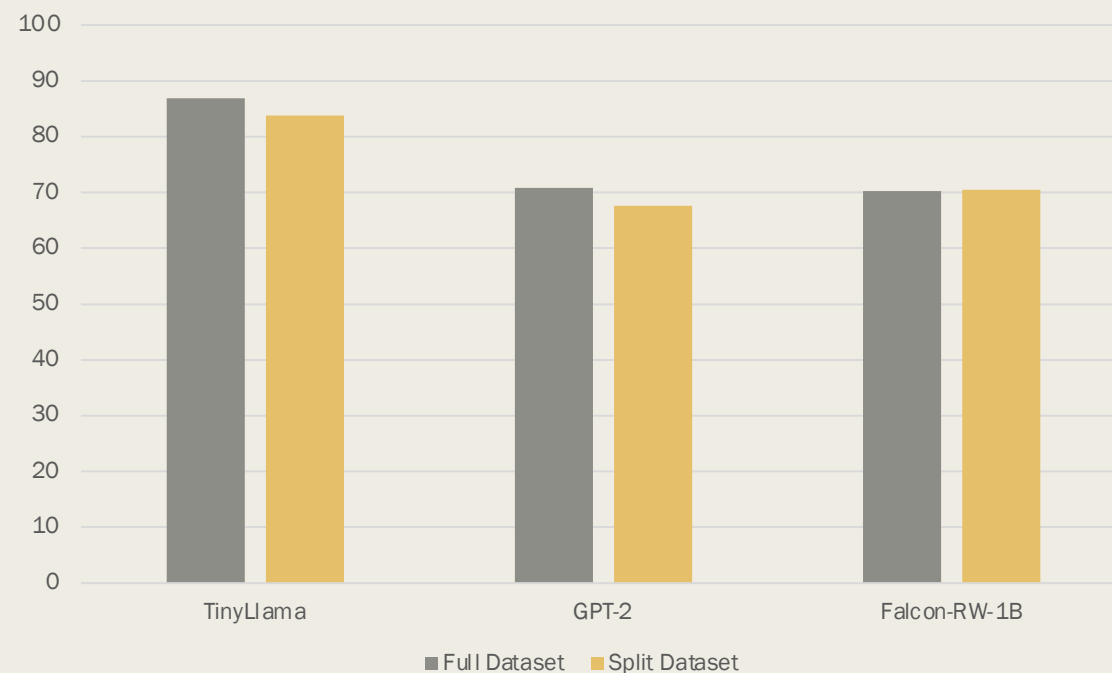


QLoRA Fine-Tuning Resource Utilization

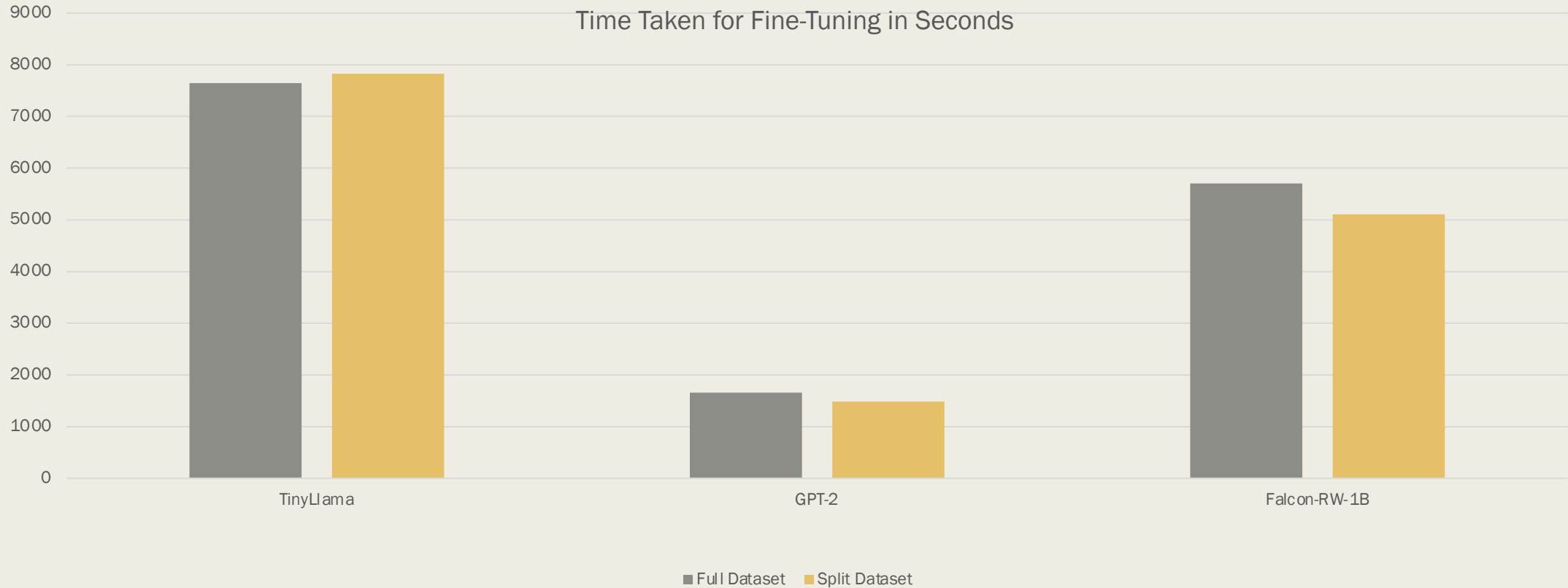
% GPU Utilization



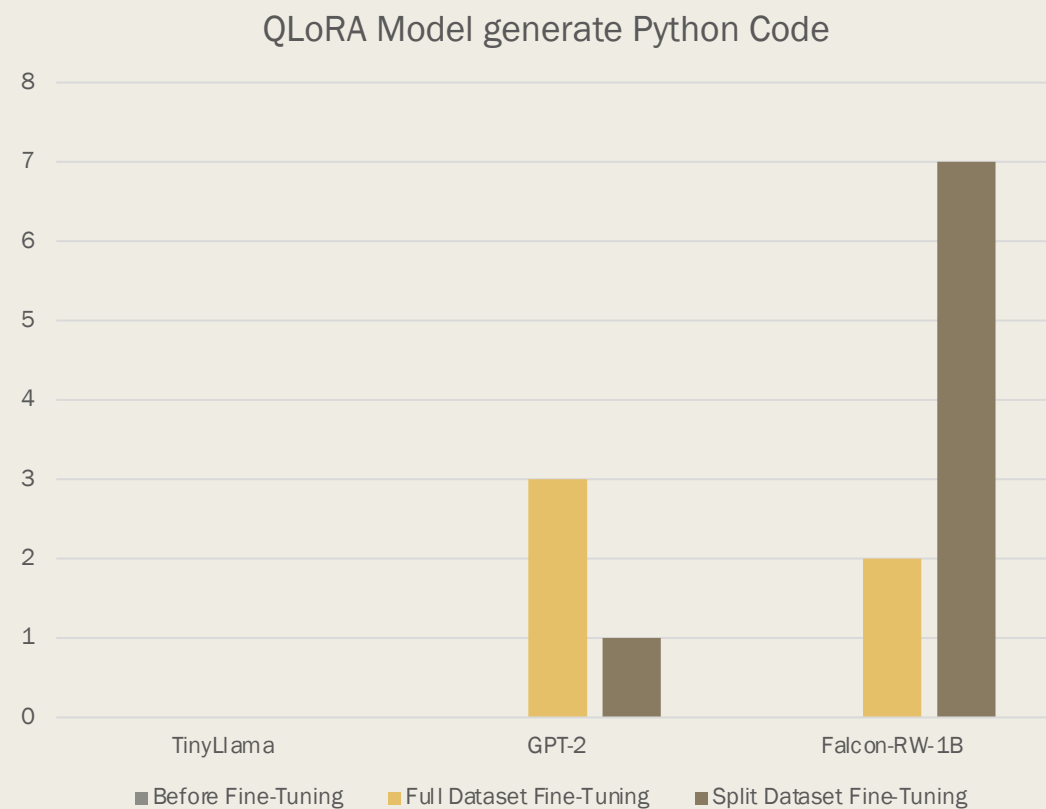
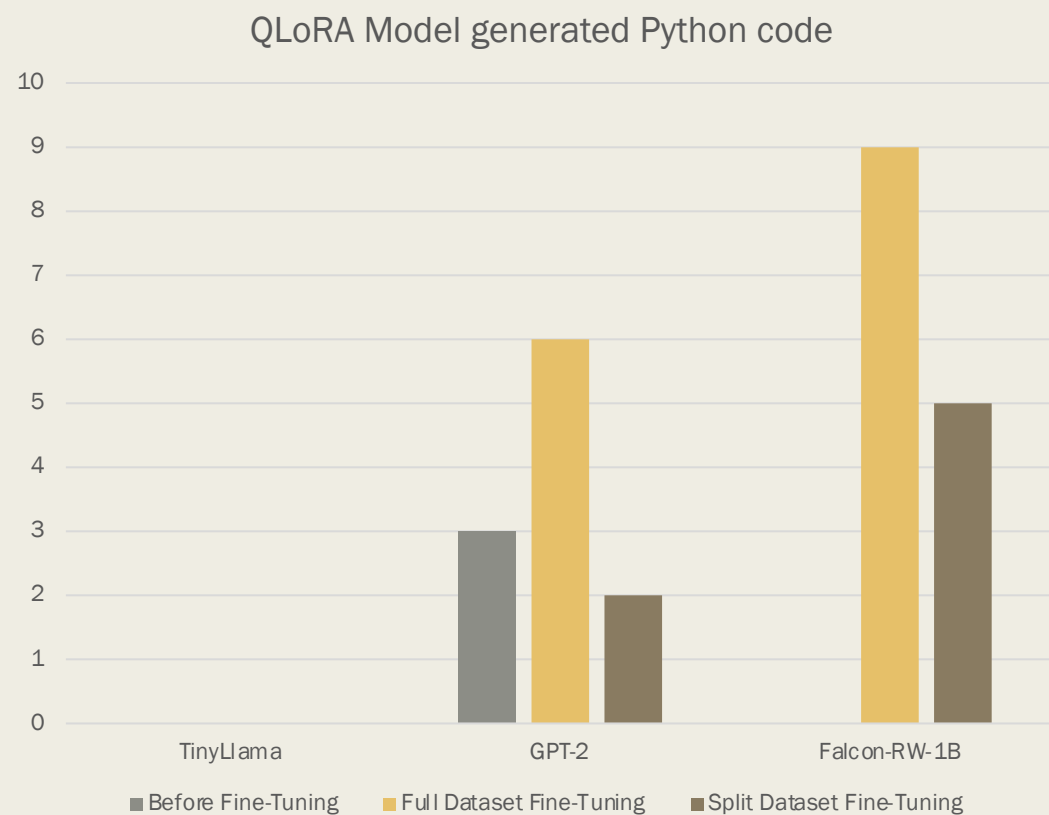
% Memory Utilization



QLoRA Fine-Tuning Time Taken



Model Inference Accuracy



Results & Conclusions

- SDFT is more memory and GPU efficient than FDFT
- SDFT is faster compared to FDFT
- FDFT showed better inference accuracy

Future Work & Recommendations

- Use better GPU
- Fine-Tune with Larger Datasets
- Explore other PEFT methods
- Hyperparameter tuning

Thank you